SAISIMRAN DATTA

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</> **Bioprocess Engineering, Vaccine Manufacturing, Biotech Commercial Strategy.**

EDUCATION

MSc Biochemical Engineering 2025 - 2026

University College London

- Averaging **76%** across all taught modules completed to date — on track for a **Distinction**. Dissertation in progress.
- Coursework centred on biotherapeutic manufacturing strategy, cost-efficiency and scale-up, taught through monoclonal antibody, biofuel and bioprocessing case studies.

BSc Microbiology 2022 - 2025

Imperial College London

- Awarded 2:1; dissertation awarded **First Class**.
- Dissertation on cerebral malaria: analysed TEER, permeability, Western blot and ELISA datasets from a triculture blood–brain barrier model to characterise protein dysregulation, and linked multivalent vaccine design to uptake, access and supply-chain efficiency in sub-Saharan Africa.
- Second research project on dynamic Foxp3 expression in autoimmune disease, also awarded **First**.

PROJECTS

ChAdOx1 Bundibugyo Ebolavirus Vaccine — Manufacturing Design 2026

MSc, UCL

- Derived a one-million-dose design basis from a 500,000-patient global stockpile at two doses per patient, then designed the end-to-end process: upstream and downstream process design, mass balances, equipment sizing, scale-up and techno-economic analysis.

Monoclonal Antibody Clarification Process Optimisation 2026

MSc, UCL

- Redesigned the clarification train for a mAb process; sizing calculations projected a **fourfold** reduction in required filtration area, with lower downstream fouling risk.

Recombinant Protein Expression & Bioprocess Data Analysis 2026

MSc, UCL

- Analysed recombinant protein expression across four induction temperatures in R, identifying 23 °C as the optimum trade-off between growth rate, yield and product quality.

STI Outbreak Micro-Simulation Model 2023

Self-directed

- Built a stochastic micro-simulation model to forecast sexually transmitted infection outbreaks across London student accommodation.

EXPERIENCE

Investment Research Intern Nov 2025 - Jan 2026

Seguro, Dubai

- Authored coverage reports on **Eli Lilly**, **Novo Nordisk** and **NVIDIA**, working from SEC filings, company annual reports and primary press coverage.
- Argued that Lilly's concentration in a single GLP-1 obesity franchise was the principal risk to the story, and read its $42.5B of debt against $9.8B of cash as the price of a $50B+ US manufacturing build-out. Made the parallel case at Novo Nordisk, where FY24 free cash flow of DKK −14.7bn reflected capacity expansion and an acquisition rather than weakening demand.
- Brought a manufacturing perspective to both pharmaceutical names: the Wegovy access shortfall came down to the specialised production that biologic injectables require, while the routine MRI screening demanded by ARIA sets a practical ceiling on Kisunla's market.
- Issued dated six- and twelve-month theses on NVIDIA, arguing that TSMC's CoWoS packaging capacity and Chinese export licensing — not unit demand — were the dominant risks.

Analyst (Part-time) Oct 2023 - Oct 2025

G.D Management Consultancy, Dubai

- Sustained a two-year part-time analyst engagement alongside a full-time BSc, evaluating growth opportunities and market dynamics across UAE client engagements.
- Prepared executive-ready presentations and briefing materials, distilling business research into recommendations for client decision-makers.

Undergraduate Research Assistant Aug 2024 - Oct 2024

Dept. of Life Sciences, Imperial College London

- Worked in Prof. Faith Osier's malaria immunology group on next-generation malaria vaccine candidates: maintained EXPI293 mammalian cell culture, ran transient transfections, and purified and characterised merozoite proteins by affinity chromatography.
- Cleaned, analysed and visualised antibody assay data from clinical samples in R, identifying drivers of cell viability that informed protocol changes reducing cell death by **30%**.
- Redesigned multi-step protocol documentation into flow-chart visual aids; the graphics were formally adopted into the lab manual for Imperial's final-year Vaccinology course.

Logistics & Regulatory Affairs Intern Jul 2023 - Oct 2023

LIFE Pharma, Dubai

- Analysed stability and compositional datasets for clopidogrel tablets and prepared a USP-aligned stability summary supporting a successful drug approval.
- Reviewed product and label data for accuracy, traceability and regulatory compliance, contributing to claims documentation and market readiness.

Quality Assurance Intern Jul 2022 - Sep 2022

LIFE Pharma, Dubai

- Quantified active pharmaceutical ingredient content in Seltaflu tablet batches by **HPLC**, verifying dosage accuracy, product quality and regulatory compliance.
- Coordinated across production, regulatory, marketing and sales functions to resolve approval requirements and protect product supply timelines.

WRITTEN WORK

Epidemic-Response Vaccine Platforms: A Landscape Analysis 2026

MSc dissertation research, UCL

- Built a comparative decision framework for filovirus outbreak response, scoring six vaccine platforms — live-attenuated, inactivated whole-virus, protein subunit, virus-like particle, mRNA and viral-vector — against five parameters: scalable manufacturability, immunological efficacy, stability, analytical complexity and deployment logistics.
- Concluded that no platform dominates across all five, and that selection is an explicit trade-off driven by the epidemiological, logistical and regulatory profile of the target outbreak — e.g. mRNA's design speed is offset by cold-chain and booster-tracking constraints in resource-limited settings.

LEADERSHIP

Student Mentor, PASS Programme 2022 - 2025

Imperial College London

- Ran a root-cause analysis of persistently low lecture attendance and implemented targeted interventions, increasing participation by **80%**.

Delegate, DUPHAT 2024

Dubai Int. Pharmaceuticals & Technologies Conference

- Debated the role of AI and government regulation in biopharmaceutical manufacturing in the Middle East; discussed MDCK cell-culture production and H5N1 pandemic preparedness with Dr. Rino Rappuoli.

SKILLS

- **Bioprocess & Engineering:** Upstream and downstream processing, mass balances, equipment sizing, process flow diagrams and P&IDs, scale-up, process validation, techno-economic analysis, GMP.
- **Laboratory:** HPLC, ELISA, Western blot, affinity chromatography, mammalian cell culture (EXPI293), transient transfection, TEER and permeability assays.
- **Data & Analytics:** R (advanced), Python (intermediate), Excel (advanced), statistical analysis, data visualisation, stochastic simulation.
- **Commercial:** Equity and market research, techno-economic modelling, evidence synthesis, executive reporting, PowerPoint (advanced).
- **Languages:** English, Hindi, Punjabi and Urdu (all fluent).